Throughput QOS

2026-03-23-14-21-09

Objective

The objective of the QoS (Quality of Service) traffic throughput test is to measure the maximum achievable throughput of a network under specific QoS settings and conditions. By conducting this test, we aim to assess the capacity of the network to handle high volumes of traffic while maintaining acceptable performance levels, ensuring that the network meets the required QoS standards and can adequately support the expected user demands.

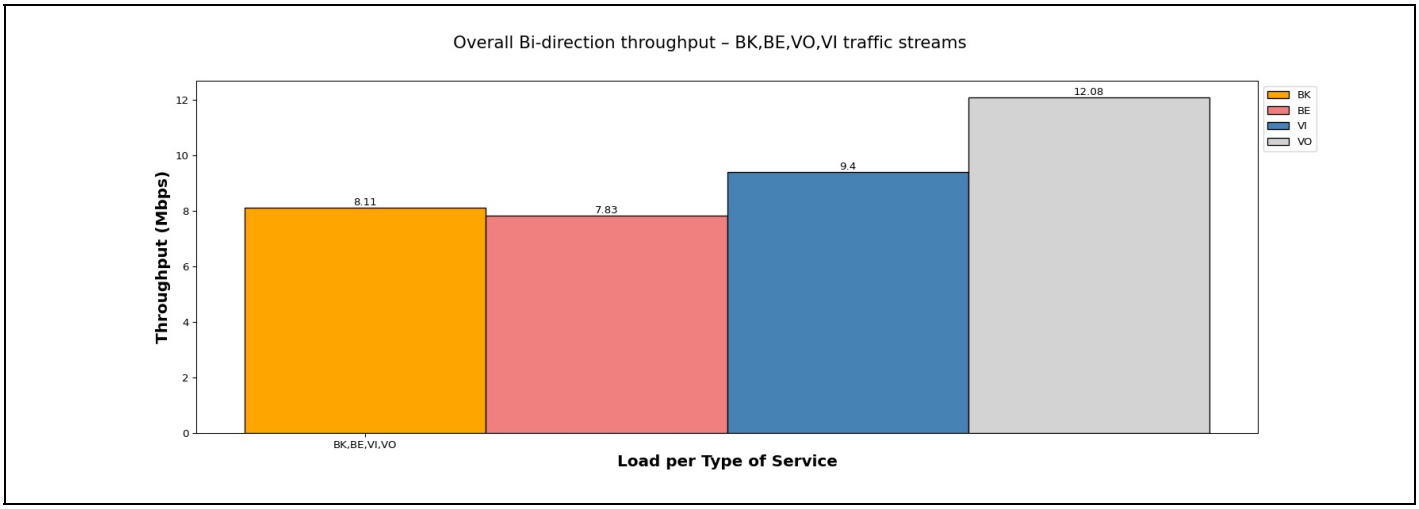
Test Configuration	Number of Stations	8
	AP Model	NETGEAR
	SSID_2.4GHz	NETGEAR_2G_wpa2
	SSID_5GHz	NETGEAR_5G_wpa2
	SSID_6GHz	None
	Traffic Duration in hours	0.02
	Security_2.4GHz	wpa2
	Security_5GHz	wpa2
	Security_6GHz	Open
	Protocol	UDP
	Traffic Direction	Bi-direction
	TOS	['BK', 'BE', 'VI', 'VO']
	Per TOS Load in Mbps	Upload:1.0 Mbps,Download:1.0 Mbps

Overall Bi-direction Throughput for all TOS i.e BK | BE | Video (VI) | Voice (VO)

No of Stations	Throughput for Load 1.0 Mbps-upload	Throughput for Load 1.0 Mbps-download
4 + 4	BK : 4.03, BE : 3.52, VI: 3.96, VO: 4.75	BK : 4.08, BE : 4.31, VI: 5.44, VO: 7.33

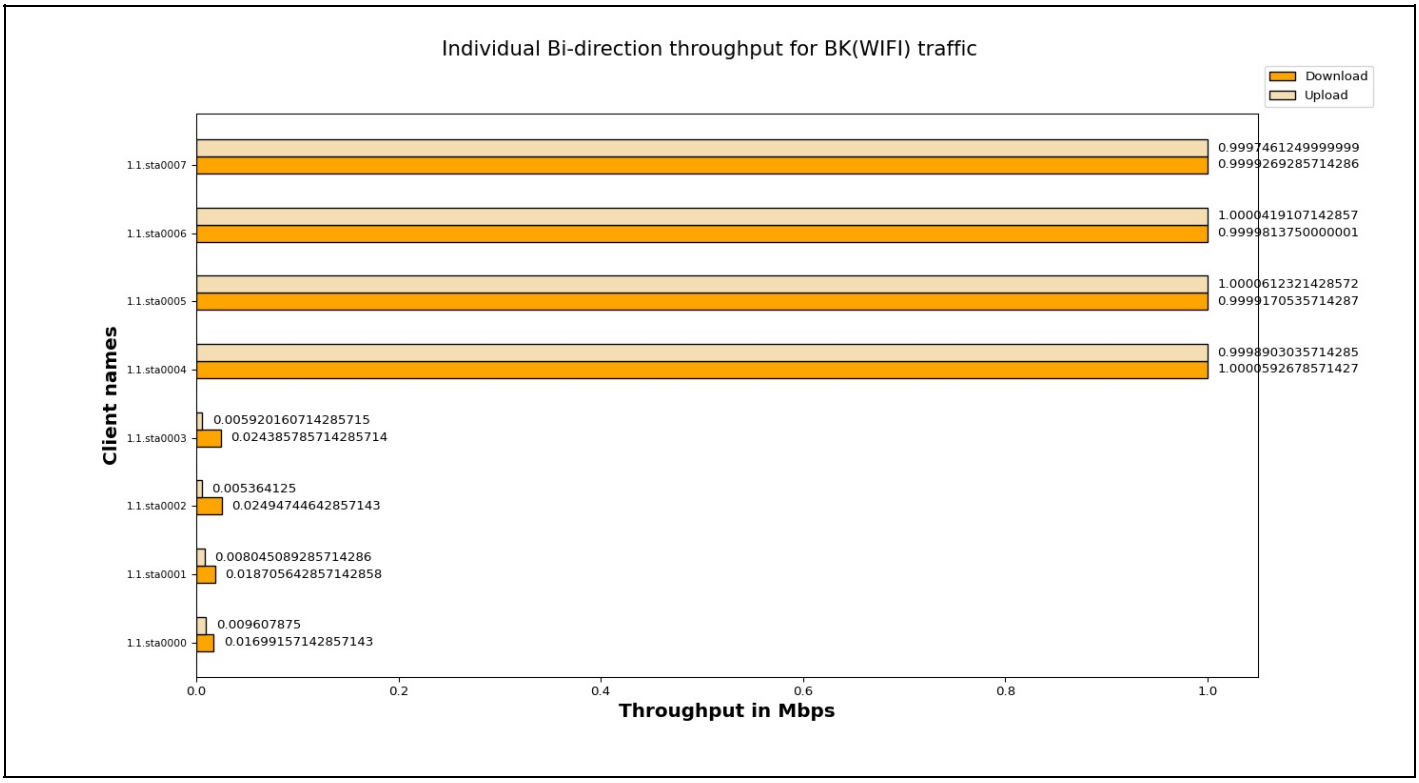
Overall Bi-direction throughput for 8 clients for 1.0 Mbps-download with different TOS.

The below graph represents overall Bi-direction throughput for all connected stations running BK, BE, VO, VI traffic with different intended loads per station – 1.0 Mbps-download



Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic BK(WiFi).

The below graph represents individual throughput for 8 clients running BK (WiFi) traffic. Y-axis shows “Client names” and X-axis shows “Throughput in Mbps”.

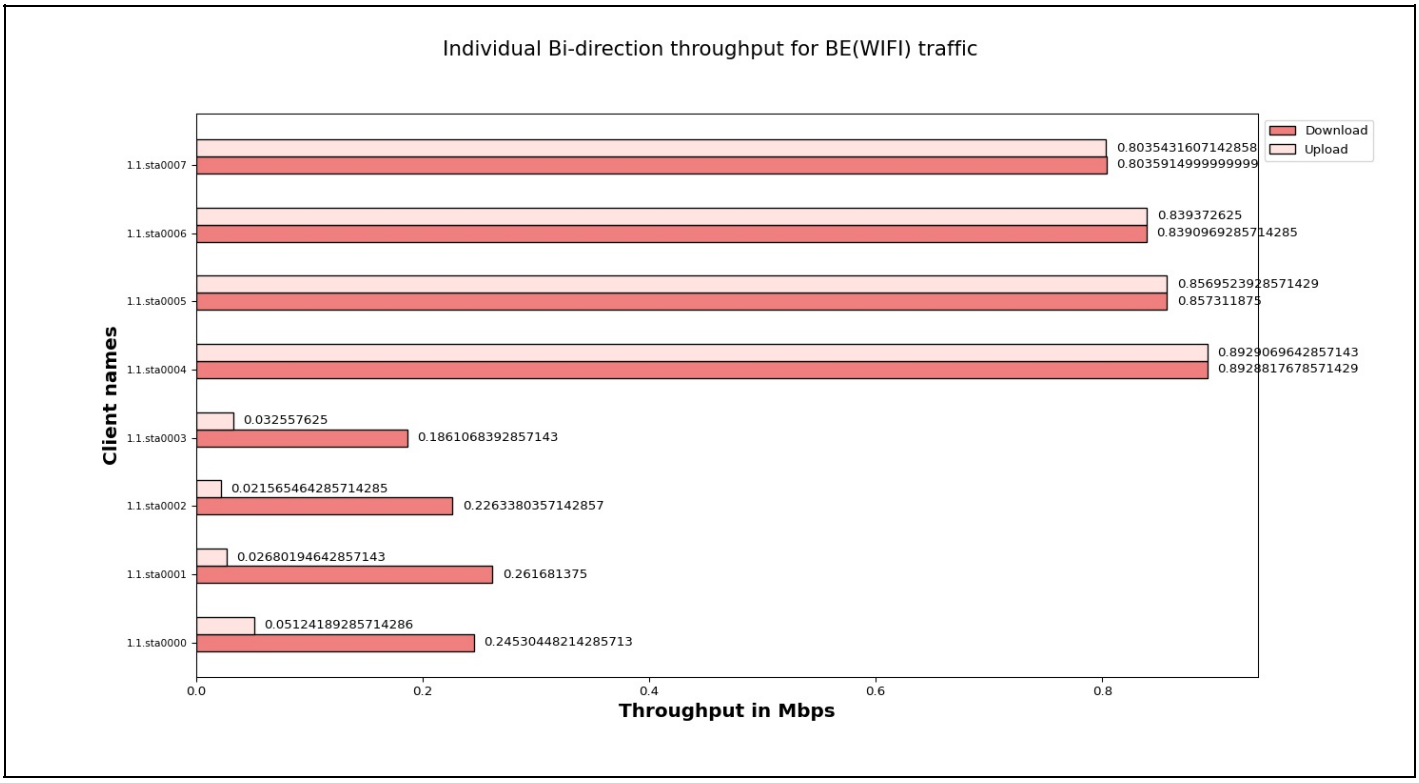


TOS : Background

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Upload Drop (%)	Observed Download Drop (%)
1.1.sta0000	04:f0:21:94:a5:46	11	-	Background	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.01 Mbps	0.02 Mbps	45.29	70.51
1.1.sta0001	04:f0:21:94:3f:46	149	-	Background	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.01 Mbps	0.02 Mbps	60.19	84.14
1.1.sta0002	04:f0:21:94:a0:46	-	-	Background	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.01 Mbps	0.02 Mbps	61.29	81.42
1.1.sta0003	04:f0:21:94:3a:46	-	-	Background	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.01 Mbps	0.02 Mbps	68.53	98.07
1.1.sta0004	00:0a:52:07:b2:c5	-	-	Background	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	1.0 Mbps	1.0 Mbps	0.01	0.02
1.1.sta0005	00:0a:52:07:50:c5	-	-	Background	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	1.0 Mbps	1.0 Mbps	0.01	0.03
1.1.sta0006	00:0a:52:07:53:c5	-	-	Background	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	1.0 Mbps	1.0 Mbps	0.0	0.02
1.1.sta0007	00:0a:52:07:3b:c5	-	-	Background	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	1.0 Mbps	1.0 Mbps	0.01	0.01

Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic BE(WiFi).

The below graph represents individual throughput for 8 clients running BE (WiFi) traffic. Y-axis shows “Client names” and X-axis shows “Throughput in Mbps”.

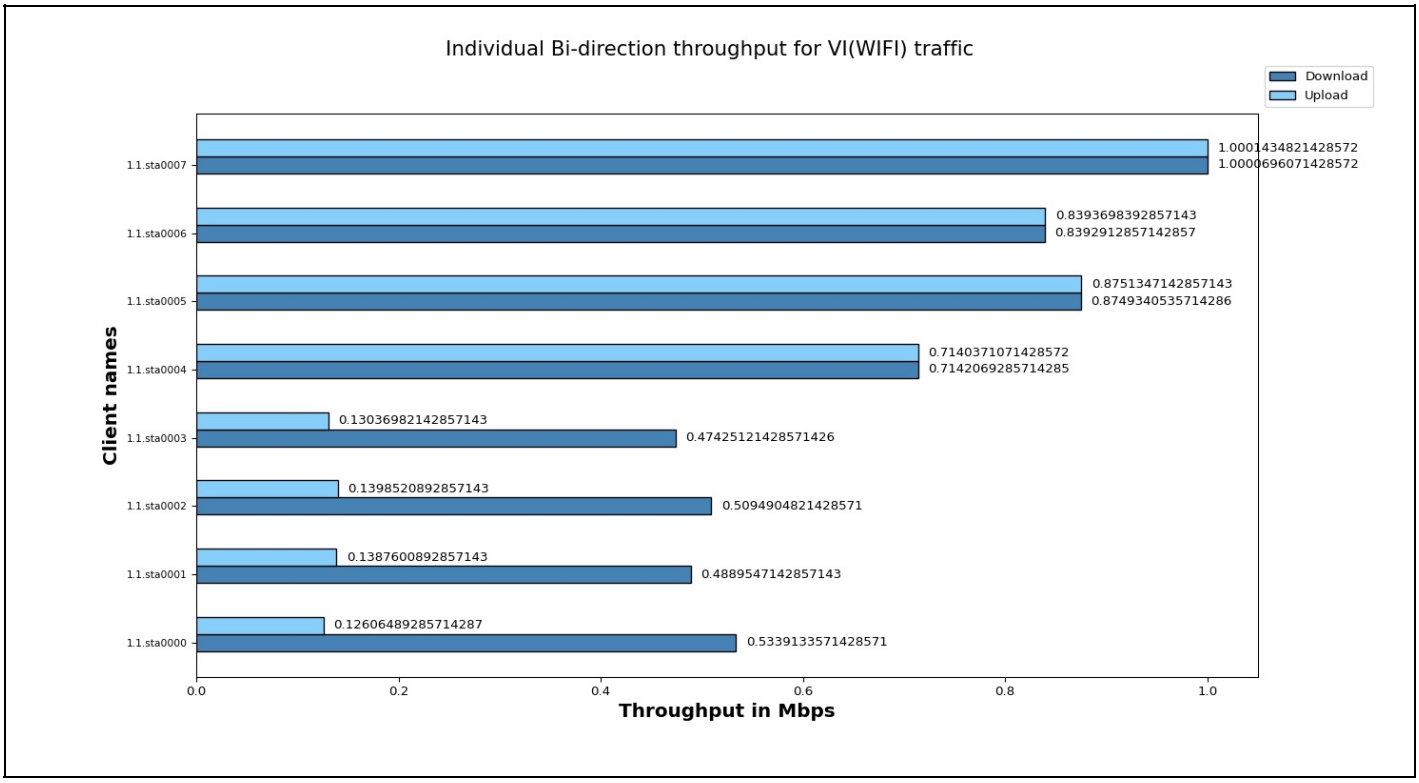


TOS : Besteffort

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Upload Drop (%)	Observed Download Drop (%)
1.1.sta0000	04:f0:21:94:a5:46	11	-	Besteffort	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.05 Mbps	0.25 Mbps	75.26	72.65
1.1.sta0001	04:f0:21:94:3f:46	149	-	Besteffort	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.03 Mbps	0.26 Mbps	78.98	77.42
1.1.sta0002	04:f0:21:94:a0:46	-	-	Besteffort	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.02 Mbps	0.23 Mbps	75.16	70.32
1.1.sta0003	04:f0:21:94:3a:46	-	-	Besteffort	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.03 Mbps	0.19 Mbps	77.91	73.11
1.1.sta0004	00:0a:52:07:b2:c5	-	-	Besteffort	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.89 Mbps	0.89 Mbps	0.0	0.0
1.1.sta0005	00:0a:52:07:50:c5	-	-	Besteffort	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.86 Mbps	0.86 Mbps	0.01	0.0
1.1.sta0006	00:0a:52:07:53:c5	-	-	Besteffort	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.84 Mbps	0.84 Mbps	0.0	0.01
1.1.sta0007	00:0a:52:07:3b:c5	-	-	Besteffort	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.8 Mbps	0.8 Mbps	0.0	0.0

Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic VI(WiFi).

The below graph represents individual throughput for 8 clients running VI (WiFi) traffic. Y-axis shows “Client names” and X-axis shows “Throughput in Mbps”.

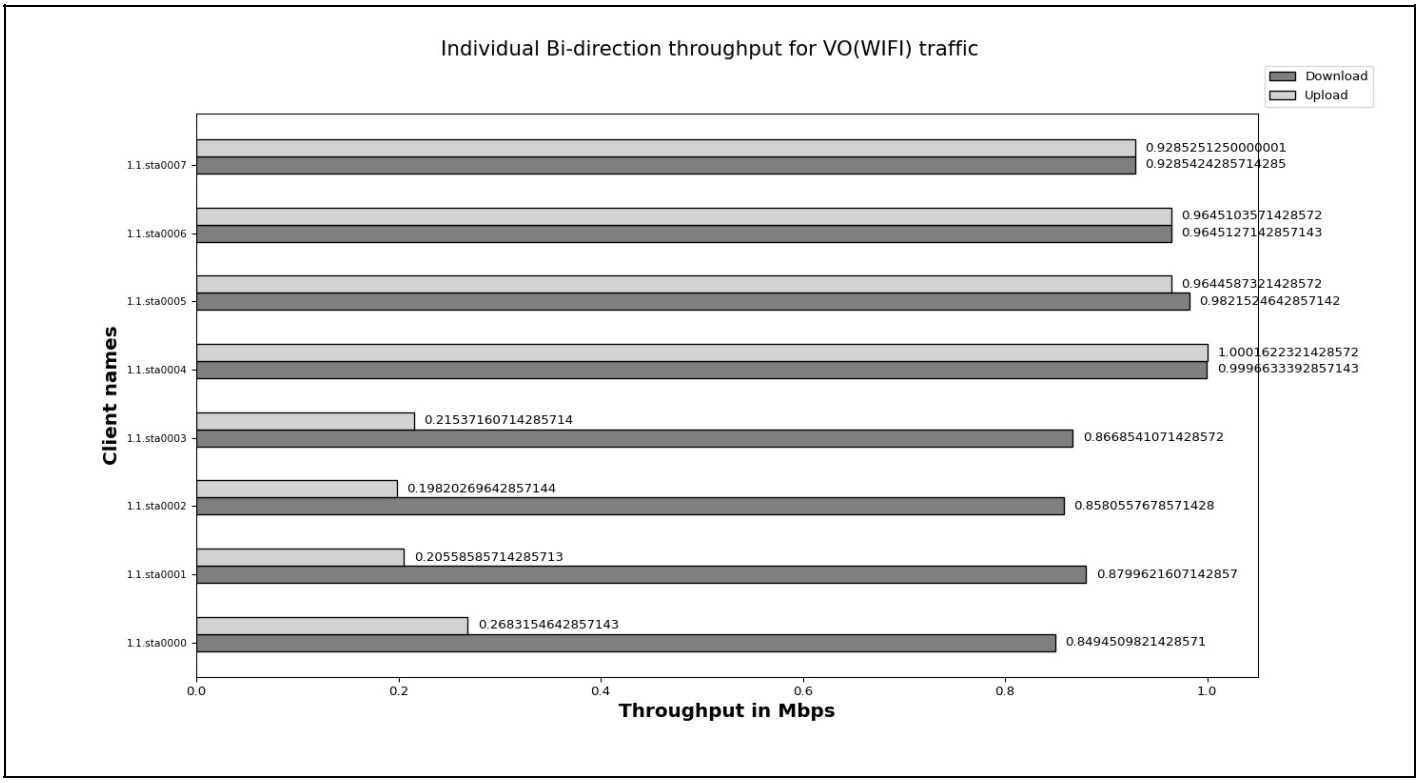


TOS : Video

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Upload Drop (%)	Observed Download Drop (%)
1.1.sta0000	04:f0:21:94:a5:46	11	-	Video	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.13 Mbps	0.53 Mbps	65.85	26.05
1.1.sta0001	04:f0:21:94:3f:46	149	-	Video	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.14 Mbps	0.49 Mbps	60.87	25.94
1.1.sta0002	04:f0:21:94:a0:46	-	-	Video	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.14 Mbps	0.51 Mbps	60.02	26.94
1.1.sta0003	04:f0:21:94:3a:46	-	-	Video	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.13 Mbps	0.47 Mbps	60.19	25.95
1.1.sta0004	00:0a:52:07:b2:c5	-	-	Video	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.71 Mbps	0.71 Mbps	0.0	0.0
1.1.sta0005	00:0a:52:07:50:c5	-	-	Video	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.88 Mbps	0.87 Mbps	0.01	0.0
1.1.sta0006	00:0a:52:07:53:c5	-	-	Video	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.84 Mbps	0.84 Mbps	0.0	0.0
1.1.sta0007	00:0a:52:07:3b:c5	-	-	Video	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	1.0 Mbps	1.0 Mbps	0.01	0.01

Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic VO(WiFi).

The below graph represents individual throughput for 8 clients running VO (WiFi) traffic. Y-axis shows “Client names” and X-axis shows “Throughput in Mbps”.



TOS : Voice

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Upload Drop (%)	Observed Download Drop (%)
1.1.sta0000	04:f0:21:94:a5:46	11	-	Voice	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.27 Mbps	0.85 Mbps	72.85	12.37
1.1.sta0001	04:f0:21:94:3f:46	149	-	Voice	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.21 Mbps	0.88 Mbps	76.77	11.93
1.1.sta0002	04:f0:21:94:a0:46	-	-	Voice	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.2 Mbps	0.86 Mbps	80.53	10.16
1.1.sta0003	04:f0:21:94:3a:46	-	-	Voice	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.22 Mbps	0.87 Mbps	78.46	12.27
1.1.sta0004	00:0a:52:07:b2:c5	-	-	Voice	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	1.0 Mbps	1.0 Mbps	0.0	1.79
1.1.sta0005	00:0a:52:07:50:c5	-	-	Voice	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.96 Mbps	0.98 Mbps	0.0	0.0
1.1.sta0006	00:0a:52:07:53:c5	-	-	Voice	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.96 Mbps	0.96 Mbps	0.0	0.01
1.1.sta0007	00:0a:52:07:3b:c5	-	-	Voice	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.93 Mbps	0.93 Mbps	0.0	0.0

Information	contact	support@candelatech.com